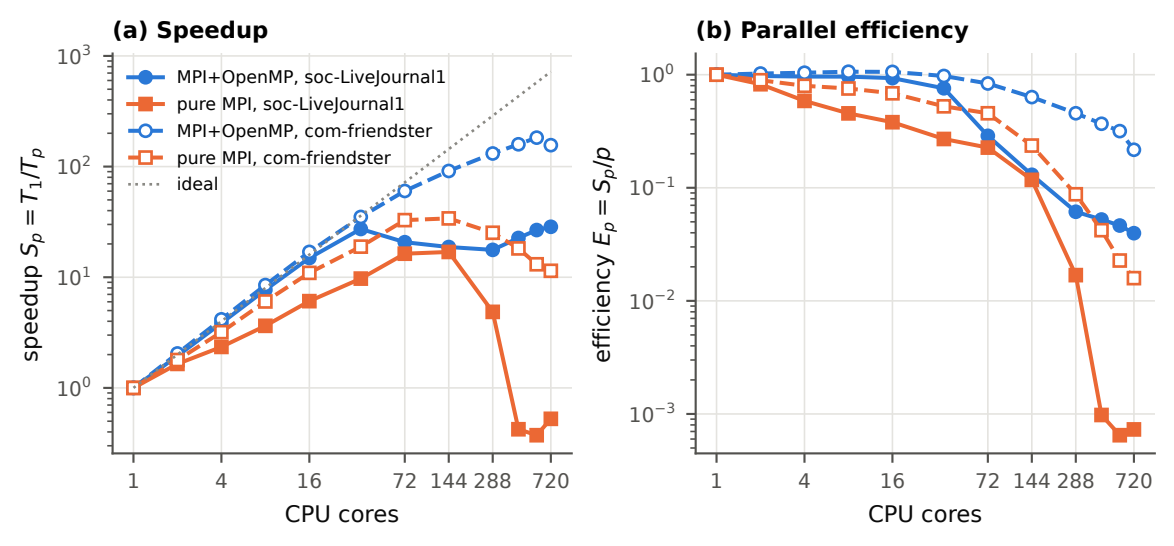
(a) Speedup
MPI+OpenMP, soc-LiveJournal1
pure MPI, soc-LiveJournal1
MPI+OpenMP, com-friendster
pure MPI, com-friendster
ideal
speedup $S_p = T_1/T_p$
10^3
10^2
10^1
10^0
1
4
16
72
144
288
720
CPU cores
(b) Parallel efficiency
efficiency $E_p = S_p/p$
10^0
10^{-1}
10^{-2}
10^{-3}
1
4
16
72
144
288
720
CPU cores